Unnamed_Unit

Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738066466071-Chapter 4_unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.

UNIT 1 EXPLORING THE CORE

Topic Investigation

WHAT ARE MUSICAL NOTES?

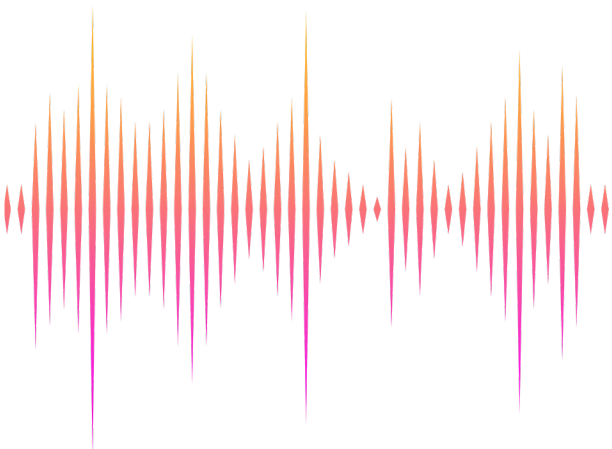
Musical notes are the foundation of any musical piece. They represent sounds at specific pitches, which when played in sequence, form melodies and harmonies that we enjoy in songs and performances. In written music, these notes are depicted using symbols placed on a staff—a series of five parallel lines. Each position on these lines or in the spaces between them corresponds to a particular musical pitch. For example, notes like A, B, C, D, E, F, and G are written on different lines or spaces, and each can be modified with sharps or flats to adjust their pitch.

HOW DO WE HEAR MUSIC?

Music reaches us through sound waves created by vibrations from musical instruments, voices, or digital outputs like speakers. When an object vibrates, it causes the air around it to move, creating waves that propagate through the environment. These sound waves enter our ears and are translated by our auditory system into sounds that our brains perceive as music. The nature of these vibrations—how fast or slow they are—determines the pitch of the sound we hear.

FREQUENCY AND PITCH

The concept of frequency is central to understanding musical pitch. Frequency refers to the number of vibrations per second and is measured in Hertz (Hz). Each musical note corresponds to a particular frequency. For example, when we say a note is "middle C," we are referring to a sound wave that vibrates at 261.6 Hz.



DETAIL ON PITCH

Pitch is our perception of how high or low a sound is and is closely related to the frequency of the sound wave. When discussing musical scales, particularly the standard Western music scale, pitch plays a crucial role in differentiating each note:

MUSICAL SCALE AND NOTES

A musical scale is a series of notes that are ordered by pitch. In the Western music system, a standard scale consists of seven notes: Do (C), Re (D), Mi (E), Fa (F), Sol (G), La (A), and Si (B). Each note on this scale is associated with a specific pitch; for instance, middle C (Do) has a lower pitch than D (Re).

FREQUENCY OF NOTES

Each successive note in a scale typically has a higher frequency than the one before it, which means each note sounds progressively higher. For example, middle C (Do) might be set at a frequency of 261.6 Hz, while D (Re) is at 293.7 Hz, and this pattern continues up the scale.

OCTAVES AND DOUBLING FREQUENCY

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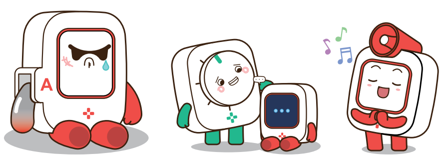

Inquiry & Reflection

Discussion question 1

Content-Related Question: Why do different instruments playing the same note sound unique? Consider the role of frequency and other factors.

Discussion question 2

Everyday Life-Related Question: How does changing the speed of a song affect its pitch and your reaction to the music?